

West Hartford Deck — Build Instructions (v15 plan)

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Scope

These instructions cover the complete construction of the 23'-6" × 16'-3" attached deck at the West Hartford property, as defined in the v15 plan and v15 Bill of Materials (BOM). The deck sits on an existing concrete patio at a maximum finished height of 12" above grade. Foundation is six Pylex 50 screw piers driven through punched patio holes; framing is three tripled 2×10 PT girders with face-hung 2×10 PT joists at 12" o.c., all topping out at 11" AG; decking is MoistureShield Vision composite with a 4-sided picture frame and MoistureShield composite fascia on three exposed sides. Two egress points: a left-edge single-step stair (42" wide, v12) and a far-edge wide step (11'-9" long, v13/v14). All hardware is ACQ-compatible HDG or stainless throughout.

How to use this document

Read the entire sequence before picking up a tool. The sections are in chronological build order — skipping ahead is how mistakes happen on a job this tight. Pay close attention to every **Don't Forget** callout marked . Those items exist because an experienced framer, working on instinct from previous jobs, would likely get them wrong on *this* deck specifically. The callouts are not generic safety reminders. They are the places where this design diverges from standard practice.

Section 0 — Pre-construction

1. Permit

Pull a West Hartford building permit before any lumber is delivered to the site. A deck of this size (≈ 380 sq ft, attached to the house) requires one — no exceptions.

Bring to the permit office: - The v15 plan (deck_plan.png or printed drawio pages — both pages, framing and decking) - The v15 BOM - The Pylex 50 load specification sheet (available on the Pylex website; shows allowable vertical load and installation method for the 50” shaft) - The Simpson DTT2Z spec sheet (lateral tie) - The Simpson HU210-3 spec sheet (girder-to-ledger connection)

The plans examiner will want to see how the house-side piers are eliminated and how the ledger carries the girder-end loads. The HU210-3 hanger + through-bolt cluster detail answers that question directly.

2. Tool list

Category	Tools
Cutting	Circular saw (with separate 40-tooth carbide blade for composite, 24-tooth for PT), miter/chop saw, reciprocating saw + bi-metal blades, jigsaw (for notch cuts and step-opening PF joints)
Driving	Cordless drill-driver, impact driver, impact wrench ($\frac{1}{2}$ ” drive for carriage bolts and lag bolts)
Drilling	Hammer drill with $\frac{1}{2}$ ” spade or auger bit for lag pilots; masonry bit + chisel set for patio punch finishing
Layout	100-ft tape + 25-ft tape (keep a second 25-ft tape — you’ll lose one), chalk line + chalk, marking spray paint (inverted), framing square, speed square, mason’s line + line levels
Leveling	4-ft level, 6-ft level, post level (for Pylex saddle-bracket plumb check), transit or laser level (strongly recommended — you’re leveling across 23 ft)
Fastening	Torque wrench ($\frac{1}{2}$ ” drive, for through-bolts at ledger clusters and pier saddle bolts), socket set + open-end wrenches, ratchet
Miscellaneous	Rubber mallet (setting saddles and tap-adjusting composite), sledgehammer (pier punching and driving), masonry chisel + 3-lb hammer (finishing punched patio holes), knee pads, sawhorses (2 pair for cut station), angle grinder +

Category	Tools
	diamond cup wheel (rental, for cleaning up pier hole edges)
PPE	Safety glasses, hearing protection, cut-resistant gloves (especially for composite edges)

3. Consumables — have these on hand before Day 1

- Chalk (blue) and marking spray paint — layout and pier marks
- Blue tape and a Sharpie — dimension references on framing before it's covered
- End-grain sealant — **Anchorseal 2 or Copper Green Wood Preservative**, 2 quarts minimum. Keep the can at your saw station; every PT cut end gets sealed the moment it's made.
- Construction adhesive — PL Premium or equivalent, for blocking glue-assist (belt-and-suspenders at butt joints)
- Shop towels and a bucket — adhesive cleanup, general
- 6-mil plastic sheeting + canvas tarp — protect staged PT lumber and MoistureShield bundles from rain overnight
- Drop cloths or construction-grade plastic — protect patio surface during demolition-phase patio punching
- Drinking water and sunscreen — this is outdoor summer work

4. Order timing

PT lumber (KDAT SYP): Call Richards Building Supply in Hartford metro first. Kiln-dried-after-treatment SYP can take 1–2 weeks to come in; standard wet-treated stock is usually on-hand but will shrink and cup post-install. Worth the wait for KDAT.

MoistureShield Vision boards: Check local stock at Home Depot, Lowe's, or Richards. The 16-ft field planks (you need 54) are typically a special-order item unless a local yard keeps full Vision inventory. Order early — you do not want framing done and decking delayed.

Cortex face-screws + plugs: Order only after you've finalized your MoistureShield Vision colorway selection. Cortex plugs are color-matched to specific product colors. Wrong colorway = conspicuous plugs. Do not order generic plugs.

Tiger Claw TC-G clips: Generally in stock at decking suppliers; 625 clips is a quantity they may need to pull from back stock. Confirm before your decking start date.

5. Inspection schedule

West Hartford typically requires four inspections on an attached deck permit:

1. **Footing/pier** — after piers are driven and grout collar is placed, before any framing sits on them. Inspector wants to verify pier depth (past 42" frost line) and location within the patio footprint.

2. **Framing / pre-decking** — after all framing (ledger, girders, joists, blocking, hangers) is complete and before any decking or joist tape goes down. **This is the critical one.** The inspector needs to see every hanger, every nail pattern, the ledger connection clusters, and the DTT2Z lateral ties.
3. **Post-decking + railings (if applicable)** — after decking and any trim/fascia is in place. This deck is $\leq 12''$ AG so no guardrail is required by IRC R312, but confirm with the West Hartford building department.
4. **Final** — project complete; inspector walks it.

Schedule the framing inspection BEFORE applying joist tape. The inspector needs to see the hanger nail patterns on bare wood. Once tape is down, that's over.

Section 1 — Site prep, patio protection, and layout

What you're doing

Protecting the existing concrete patio from construction damage, establishing the precise deck footprint in the field, and confirming squareness before a single pier hole is punched. Errors here propagate through the entire build; errors in layout are free to fix at this stage.

Sequence

1. Lay construction-grade 6-mil plastic across the entire patio surface. Overlap seams by 12". Tape all edges to the house foundation. Cover plastic with canvas tarp if rain is in the forecast or if heavy materials will be staged on it.
2. Mark the house-wall ledger line at the correct elevation: top of decking will be 12.0" AG; subtract 1.0" (MoistureShield Vision thickness) and 9.25" (2×10 actual depth) = ledger top at 1.75" AG. Snap a chalk line on the house wall at this elevation across the full 23'-0" ledger run.
3. Mark all six pier center locations:
 - **Middle row:** x = 0', 11'-6", 23'-0" at y = **8'-1.5"** from house
 - **Outer row:** x = 0', 11'-6", 23'-0" at y = **16'-0"** from house
 - Use inverted spray paint to mark a dot at each center; put the chalk line intersection in the center of the dot.
4. Confirm squareness: measure the diagonal of the proposed deck footprint (corner to corner in both directions). Both diagonals must be equal (3-4-5 triangle check on each corner first, then diagonal-equal overall). If they aren't equal, shift layout before punching anything.
5. Confirm all six pier marks fall within the 23'-6" × 16'-6" patio footprint. Measure from the patio edges, not from your chalk lines. If any mark is at or past the patio edge, stop and reassess.

Don't Forget — specific to this deck

-  **The outer pier row is at y=16'-0", not y=16'-3".** If you have any v13 marks on the patio surface (y=16'-3"), erase and re-mark. Punching the old marks puts girders in the wrong location and the 16-ft stock won't reach.
-  **There are only two pier rows — 6 piers total.** No house-side pier row at y=12". Any framer who has built a deck before will want to put a row near the

house. The ledger carries those loads via the HU210-3 hanger clusters. Do not add piers.

-  **The far edge of the deck (PF outer face) is at y=16'-3", not y=16'-6".** The patio extends to y=16'-6". A 3" strip of bare concrete will be visible along the far edge — this is intentional and documented. Do not try to stretch the framing to cover it.

Inspection / quality check before moving on

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All six pier marks are spray-painted on the patio surface

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Diagonal-equal check passes ($< \frac{1}{4}$ " difference)

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All six pier marks confirmed to be inside patio footprint by tape measurement from patio edge

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Plastic + tarp protection is fully in place; no bare patio exposed under staging area

Section 2 — Pylex pier installation

What you're doing

Punching six holes through the concrete patio and driving Pylex 50 screw piers below the 42" frost line. Each pier receives a saddle bracket set to the correct elevation to carry the tripled girders.

Sequence

1. Set up the angle grinder with a diamond cup wheel. Score a 4" diameter circle around each pier mark to prevent slab cracking when you sledgehammer the center.
2. Use the cold chisel + 3-lb hammer to open the slab at each mark; follow with the sledgehammer to knock through. Keep the hole tight — a loose or oversize hole makes grout-collar work harder.
3. Drive each Pylex 50 pier at the punched hole location using the prescribed Pylex installation method (rotation with a pipe wrench or breaker bar). Drive until the helix tip clears 42" below the top of the slab (Pylex 50 shaft is 50" — you should have ~7–8" of shaft above the patio surface after driving to frost depth).
4. Install the Pylex saddle bracket on each pier shaft top. Bolt with $\frac{1}{2}$ " HDG carriage bolts per Pylex spec (2 per saddle). The saddle top must be set at the correct elevation: **top of decking (12") minus 1.0" decking minus 9.25" joist = saddle top at 1.75" AG**. Use a transit or laser level to verify all six saddle tops are at the same elevation before tightening down. Shim with cedar or HDPE shims where the patio surface is slightly uneven.
5. Plumb-check each pier shaft with a post level before final tightening.

6. Pack hydraulic cement around each pier shaft at the patio surface (the grout collar). Keep the collar flush with or slightly below the patio surface so it doesn't create a trip lip. Let cure per product directions before loading.
7. Once cured, touch up the surface cosmetically with Quikrete Vinyl Concrete Patcher around each collar.

Don't Forget — specific to this deck

-  **All six piers must land within the patio footprint.** If your punch drifts outboard — even by a few inches — the pier tip will be in native soil, not locked in concrete at the surface. Measure from the patio edge after punching each hole before driving. If a hole is out of position, patch it and re-punch in the correct location.
-  **Saddle elevation is more important than saddle level in isolation.** All six saddles must be at the same elevation (+/- 1/8"). If one is high or low, everything hung from it will be off. Check with the transit or laser after each installation, not just visually.
-  **Call for pier inspection before framing.** Inspector must verify pier depth. Don't put girders on piers that haven't been inspected.

Inspection / quality check before moving on

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All six piers driven; helix tips confirmed below 42" from patio surface

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All six saddle tops at 1.75" AG per laser level check ($\pm 1/8"$)

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Grout collars placed and cured at all six pier shaft penetrations

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Pier inspection passed

Section 3 — Ledger installation, flashing, and lateral ties

What you're doing

Installing the 2×10 PT ledger on the house wall and flashing it correctly. The ledger is the only house connection on this deck. Do it right the first time — fixing a leaky ledger after framing is done is a major operation.

Sequence

1. Carefully cut the siding along the ledger run. Slot the siding at the top of the cut so that the top leg of the Z-flashing can slide behind it. Keep the siding cut tight — no wider than it needs to be.
2. Apply peel-and-stick flashing (Henry Blueskin VP160 or Carlisle CCW-705, 6" wide) to the sheathing/housewrap behind the ledger location. Lap it up the wall

- at least 4" above the top of where the ledger will sit; lap it down and over the bottom of the cut sill. Work from bottom to top so each strip overlaps the one below.
3. Set the 2×10 PT ledger at the marked elevation and tack it temporarily. Confirm top elevation with level across the full 23'-0" run.
 4. Fasten the ledger using the v15 spec:
 - **At each of the three girder-to-ledger meeting points (x=0', x=11'-6", x=23'-0"):** install the Simpson HU210-3 hanger first (face-nail per spec), then drive the six ½" × 6" HDG carriage bolts per cluster. A helper must be inside the basement holding the backup block (2×10 PT scrap) against the interior face of the band joist while you drive bolts through. Do not skip the back-blocking.
 - **Between the three hanger clusters:** ½" × 4" HDG lag bolts at 18" o.c., two-row staggered per IRC R507.9.1.3. Pre-drill every lag hole with the ½" spade or auger bit.
 5. Set the Z-flashing cap over the ledger top. Slide the top leg up behind the siding (the cut you made in Step 1). The top leg goes under siding, not over it. **No caulk on the top leg or at the siding/Z-flashing interface.**
 6. Install kick-out flashing at both ends of the ledger where siding terminates. Kick-out diverts water away from the house rather than letting it run down behind the ledger end.
 7. Install Simpson DTT2Z lateral ties per IRC R507.9. These connect the deck band joist through the ledger to interior floor framing. The basement is accessible — this is an easy installation here. Location: space DTT2Z devices evenly along the ledger, staying clear of the HU210-3 hanger cluster zones. Confirm exact placement with the building inspector at time of framing; interpretation varies by jurisdiction.

Don't Forget — specific to this deck

-  **Flashing before ledger, not after.** The peel-and-stick goes on the sheathing first, then the ledger goes up. If you mount the ledger and try to stuff flashing behind it, you'll have gaps that will leak.
-  **Use Simpson-specified fasteners for the HU210-3.** The spec sheet for this hanger calls for SDS ¼"×1½" screws or N10 nails — not 16d commons. The wrong fastener can mean the hanger doesn't achieve its rated load. Read the spec sheet before opening a box of nails.
-  **Through-bolts at the HU210-3 cluster locations need a helper inside the basement.** Back-blocking must be in contact with the interior face of the band joist before the bolt goes through. Don't try to do this solo.
-  **No caulk at the Z-flashing top leg / siding interface.** Caulk here seals the drainage path shut. In CT freeze-thaw cycles, caulk at this joint fails within 3–5 years and traps water that can't escape. Leave it free to drain.
-  **Ledger fastener density is NOT uniform.** The three girder-to-ledger intersections get carriage-bolt clusters (6 bolts per cluster). The spans between get standard IRC lag pattern. This asymmetry is intentional — the ledger is now carrying point loads from three girder ends, not just distributed load.

Inspection / quality check before moving on

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Peel-and-stick flashing applied to sheathing; top lap confirmed under siding cut

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- ☐ Z-flashing top leg slides freely behind siding; no caulk at interface
 - ☐ Kick-out flashing at both ends of ledger
 - ☐ Ledger top elevation confirmed level across full 23'-0" run
 - ☐ Three HU210-3 hangers installed at x=0', 11'-6", 23'-0" with correct fasteners
 - ☐ Through-bolt clusters installed (6 per cluster) with back-blocking in place
 - ☐ IRC lag pattern installed between clusters
 - ☐ DTT2Z lateral ties installed
 - ☐ Ledger elevation matches saddle elevations (both at 1.75" AG top of framing member target)
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Section 4 — Girders

What you're doing

Building and setting three tripled 2×10 PT girders, each 16'-0" long. These span from the ledger (house end) across both pier rows to the outer rim of the deck.

Sequence

1. Lay out the three plies for Girder 1 on sawhorses. Confirm each ply is 16'-0" net length (measure it). Stack them crown-up (crown = the slight natural bow along the board's length, convex side up) so all three crowns go the same direction, then face-fasten with GRK RSS or SDS ¼"×3" structural screws staggered top/bottom/middle at ~16" o.c. down the run. Alternate: through-bolt with ½" × 3½" HDG carriage bolts at ~24" o.c. per AWC 3-ply spec.
2. Repeat for Girder 2 and Girder 3.
3. Set Girder 1 (left, x=0') into the HU210-3 hanger at the ledger and onto the two saddle brackets (middle row and outer row piers). With a helper, hold it level and confirm the saddle fit. Seat it fully into the saddle before nailing the hanger. Do not fully tighten the saddle lag bolts until level is confirmed.
4. Set Girder 2 (mid, x=11'-6") and Girder 3 (right, x=23'-0") the same way.
5. With all three girders set, check level along each girder run AND across the 23'-ft direction. Use the transit or laser — eye-balling is not adequate over 23 ft. Shim at saddle brackets as needed to achieve level across all three girder tops. This is the surface everything else references.
6. Tighten saddle bolts to snug-plus-quarter-turn (torque wrench). Seat ledger-end HU210-3 hanger nails per spec.

Don't Forget — specific to this deck

- **The 16-ft ply length is exact — use the full board with no cuts.** One of the main improvements in v14 was eliminating the v13 16'-3" ply (which needed 18-ft boards or yard-override stock). In v15, you order standard 16-ft 2×10 PT and install it full length. If you or the yard are tempted to trim anything, stop and re-measure. All three girders should span from ledger face to outer pier saddle with no cut ends.
- **All three girder tops must be co-planar and at 11.0" AG.** This is the single most important elevation in the build. Every joist, every piece of decking, the 12" AG ceiling — all derive from here. Use the transit or laser to confirm before nailing anything home.
- **Seal every cut end immediately.** Even if you're not cutting ply length, you may cut blocking or trim pieces from this same lumber. Seal cut ends with Anchorseal or Copper Green as each cut is made. PT lumber's protection is the treatment profile; a bare cut end is unprotected.

Inspection / quality check before moving on

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All three girders are 16'-0" ply length — confirmed by measurement

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Girder tops level along each run ($< 1/8"$ variation over 16 ft)

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Girder tops co-planar across the 23-ft direction ($< 1/8"$ variation)

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All saddle bolts torqued; HU210-3 hanger fasteners fully set

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All cut ends sealed

Section 5 — Joists, blocking, and joist tape

What you're doing

Hanging 36 joist segments (two per row $\times$ 17 rows + doubled far-side rim = 36 total segments) face-hung at 12" o.c., parallel to the house wall. Adding two rows of solid blocking for lateral stiffness. Then taping every framing top before any decking touches it.

Sequence

1. Mark the joist layout on the top face of all three girders: start at $y=12"$ (1'-0") from the house-wall face for the first row, then mark at 12" o.c. through $y=16'-0"$ for the 16th regular row. Mark the doubled far-side rim at $y=16'-3"$ (the outer rim).
2. Install Simpson LUS210Z hangers on the girder faces per the layout marks: one at each end of each joist segment. All hangers face-mount to the girder's inside face; top of hanger is flush with top of girder.

3. Cut joist segments from 12-ft stock to the clear span (~10'-11¼" between girder faces). Order of cuts: measure each bay independently after girders are set — don't assume all bays are identical. Seal every cut end.
4. Set each joist segment into its pair of LUS210Z hangers and nail with the specified fasteners (N10 or SDS ¼"×1½" per Simpson LUS210Z spec sheet). Use a speed square to confirm each joist is perpendicular to the girder before nailing.
5. For the far-side rim: install LUS210Z hangers for both plies at the y=16'-3" position. Hang the inner ply first; then hang the outer ply tight against it. Through-bolt the two plies together per the v15 BOM.
6. Install left and right drop-board sub-fascia: one 2×10 PT 16'-0" board on the outboard face of each side girder. These boards are at joist elevation and provide full backing for the 3" PF cantilever on the left and right sides. Lag-bolt or structural screw to girder outer face.
7. Install the far-side drop-board sub-fascia (23'-0" length) on the outboard face of the far-side doubled rim. Same purpose — backs the far-side PF cantilever.
8. Install field blocking (two rows): Row 1 at x=5'-9", Row 2 at x=17'-3". These run between joist segments perpendicular to the joist direction (i.e., blocking runs parallel to girders). Cut from 12-ft 2×10 drop-offs; face-nail through joist faces or use Simpson face-mount connectors. Apply a bead of PL Premium on the blocking-to-joist contact face before nailing.
9. Install house-side PF backing blocks: short 2×10 PT pieces between the ledger face and the first joist row (y=0 to y=12"), at ~12" o.c. Nail or screw to ledger face. These back the house-side picture-frame board.
10. **Call for framing inspection.** Inspector must see all hangers, nail patterns, lateral ties, and ledger connection before you apply joist tape.
11. After inspection passes: apply Vycor Deck Protector or Trex Protect 2½" tape to the top face of every joist segment, every girder top, and the full run of the ledger top. Apply tape the full length of each member; butt-joint and overlap tape at intersections. Press firmly; tape must lay fully flat.

Don't Forget — specific to this deck

-  **Joists run PARALLEL to the house — along the 23'-6" direction.** This is the most common misread on this deck. On a typical attached deck, joists run perpendicular to the house wall (into the yard). On this deck the joists run sideways (parallel to the house) because the girders run into the yard instead. If a framer instinctively starts laying out joists perpendicular to the house, stop the work.
-  **12" o.c. joist spacing, not 16".** The homeowner spec'd 12" for stiffness under composite decking. This means more joist segments, more LUS210Z hangers, and a tighter spacing than a framer may default to. Confirm your layout marks before you start hanging hangers.
-  **LUS210Z and HU210-3 require different fasteners.** Do not mix them up. The HU210-3 fastener holes are larger and may look the same in the box; the nail spec is different. Read the spec sheet on each box before loading your gun.
-  **Joist tape goes on framing TOPS ONLY after framing inspection passes.** Do not tape before the inspector signs off — you'll have to peel it to expose the hangers. Do not skip taping the ledger top and the girder tops — water pools at the ledger-joist intersection and rots framing from the top down if that surface isn't protected.

-  **Joist tape before decking — this is not optional.** MoistureShield Vision warranty and good practice both require protection of framing tops from moisture accumulation. Tape before the first decking plank is set.

Inspection / quality check before moving on

- ☐ All LUS210Z hangers installed with correct nail/screw specification
 - ☐ Joist spacing confirmed at 12" o.c. across the full run
 - ☐ Joist tops flush with girder tops at all hanger points (no crowning, no gaps)
 - ☐ Drop-board sub-fascia installed on all three exposed sides (left, right, far)
 - ☐ House-side PF backing blocks installed at ~12" o.c.
 - ☐ Field blocking installed at x=5'-9" and x=17'-3"
 - ☐ Framing inspection PASSED — inspector signed off before tape went on
 - ☐ Joist tape applied to all joist tops, both girder top runs, and ledger top — fully pressed down
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Section 6 — Decking, picture frame, and fascia

What you're doing

Installing the 49 field planks with Tiger Claw TC-G hidden fasteners, the 4-sided picture frame, composite fascia on three exposed sides, and Cortex face-screws + plugs at all face-fastened locations.

Sequence

1. Snap chalk lines on the joist tops for the picture frame inboard edge (2.4" inboard of the rim outer face on all four sides). The field planks land inside this line; the PF boards land outside.
2. Install the house-side picture-frame board first. It butts the wall with a ¼" thermal gap, and sits on the PF backing blocks behind it. Pre-drill every Cortex hole. Drive Cortex face-screws and set the color-matched plugs flush using the Cortex driver tool.
3. Install the far-side PF board (mitered at both outside corners, left and right). Set in position but don't fasten until the right-side and left-side PF boards are dry-fit — you need all three mitered outside corners to close correctly before committing.
4. Install left-side and right-side PF boards (mitered at the far outside corner; butt-end at the house). **Note the left-side PF is in two pieces:** a house segment (6'-0" from house to step opening) and an outer segment (6'-9" from outer edge

- of step opening back toward house) — the step opening (y=6'-0" to y=9'-6") interrupts the continuous run. Pre-drill both pieces before installing.
5. Once all four PF boards are dry-fit and mitered corners close cleanly, fasten with Cortex face-screws and plug flush.
 6. Install the three composite fascia boards (MoistureShield Vision, 0.67" thick, 11.25" tall): far side first (full 23'-0" run), then left side (16'-0", interrupted at step opening same as PF), then right side (16'-0", full run). Fasten fascia to drop-board sub-fascia and rim with #10 × 2½" stainless trim screws at ~3 screws per lineal foot.
 7. Begin field planks. Start from one PF board and work toward the opposite PF board. Set the first plank with consistent gap off the PF inboard face. Use Tiger Claw TC-G clips at every joist: one clip per joist per plank gap. The TC-G clip drives with a #2 Robertson (square) bit; have 5+ extra bits — composite decking chews through bits on a long run. Use 1/8" spacer gauges (paint-stir sticks work) to maintain consistent gaps.
 8. Every field plank is ~15'-7.2" cut from 16-ft stock. Cut all planks before starting — batch-cut is faster than cutting one at a time. **Pre-drill at any face-screw location; MoistureShield Vision composite will split if you drive without a pilot hole.**
 9. Stagger end joints if any planks need splicing (they shouldn't on a 16-ft run — a 16-ft board is slightly longer than needed, so each plank is one piece with no end joint in the field).
 10. Set the last field plank, then check that the gap between last plank and the far PF inboard face is within MoistureShield installation tolerances. A slight gap variation on the last bay is normal and expected.
 11. Cut around any conduit sleeves or other penetrations that were rough-in'd under the decking.

Don't Forget — specific to this deck

- **Field planks run perpendicular to the house — into the yard, from house side to far side.** This is correct because the joists are parallel to the house. The planks bridge across the joists. Do not run planks parallel to the house.
- **The 3" PF cantilever past the rim is intentional on three sides.** The PF outer edge is 3" past the rim on the left, right, and far sides. Do not cut the PF flush to the rim or flush to the fascia face — the overhang is the designed finish condition.
- **Cortex plug colorway must match the selected MoistureShield Vision color.** If you're on-site and discover the Cortex boxes are the wrong color, stop and reorder. Generic brown plugs on a gray Vision deck are permanently visible.
- **Pre-drill every Cortex face-screw hole in the PF boards.** Capped composite does not forgive an undrilled screw — the cap layer delaminates or the board splits. This is the step most crews skip when they're moving fast.
- **Maintain MoistureShield's specified gap between field planks.** The Vision line spec is typically ⅛" for short lengths; check MoistureShield's current installation guide for the exact requirement for your plank length. Don't butt planks tight — composite expands; tight-butt planks buckle.

Inspection / quality check before moving on

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All 49 field planks installed, gaps consistent per MoistureShield spec

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- ☐ Four-sided PF installed; all mitered corners closed with no visible gap
 - ☐ PF cantilever past rim confirmed at 3" on all three exposed sides
 - ☐ Cortex plugs all set flush; color matches colorway
 - ☐ Fascia installed on three exposed sides; secured at ~3 screws per lineal foot
 - ☐ No field planks or PF boards missing pilot holes at face-screw locations
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Section 7 — Left-edge stair (v12)

What you're doing

Installing the 42" wide single-step stair at the left edge of the deck, with its opening spanning y=6'-0" to y=9'-6" from the house wall. Three PT 2×10 cut stringers attach to the outside face of the left girder; stringer bases rest on precast pavers over a compacted gravel pad.

Sequence

1. Confirm that the left-side PF and left-side fascia are correctly interrupted at y=6'-0" and y=9'-6". The boards should have been cut and installed in two segments (Section 6) — verify the cuts are clean and the step opening is clear.
2. Prepare the stringer landing area: excavate slightly outside the patio edge (the stringers project ~16" outboard of the left girder face), level the area, compact a 4" bed of #57 crushed stone, and set four 12"×12" precast pavers in a 2×2 arrangement as the landing pad. Check the pavers for level.
3. Cut three PT 2×10 cut stringers from 8-ft stock. Each stringer is a short cut-stringer for a single 6" rise. The top of each stringer mounts to the outside face of the left girder; the stringer descends 6" to a point 10.8" (two 5.4" planks) out from the girder face (the tread depth), then flat to the landing.
4. Install Simpson LSCZ skewable hangers on the left girder outer face at y=6'-0", y=7'-9", and y=9'-6" at the correct elevation.
5. Set each stringer into its LSCZ hanger and nail per Simpson LSCZ spec. Confirm the stringer base lands on the precast pavers at grade.
6. Install Simpson MSCZ or rigid strap anchors at each stringer base to secure the bottom of the stringer to the paver. Do not skip these — an unsecured stringer base will rock and split under use.
7. Install the two tread planks (2 × MoistureShield Vision 5.4" × 42", cut from field plank drops). Two planks side-by-side form the 10.8" tread at 6" AG. Fasten with Cortex face-screws + plugs; pre-drill.

Don't Forget — specific to this deck

-  **The left-side PF interruption must be correct before the stair goes in.** The PF boards and fascia are in two pieces at this edge: a 6'-0" house segment and a 6'-9" outer segment. The step opening between them is structural

clearance — there is no board bridging that 3'-6" gap. Verify both segment ends are square and the gap is correct before mounting stringers.

- **Stringer-to-girder fasteners are structural — use HDG structural screws.** The LSCZ hanger nails are specified on the Simpson spec sheet. The stringer-to-paver anchors are Simpson MSCZ or equivalent rated for this connection. Deck screws or drywall screws are not acceptable at any load-bearing connection on this stair.
- **The three stringer positions are y=6'-0", y=7'-9", and y=9'-6" — not evenly spaced.** The outside stringers land at the cut-joint edges of the step opening in the PF/fascia; the middle stringer is at the midpoint. Confirm positions before cutting hangers.

Inspection / quality check before moving on

☐

Left-side PF and fascia interruption confirmed at y=6'-0" and y=9'-6"

☐

Gravel pad and precast pavers set level; pavers stable under foot pressure

☐

Three stringers installed at y=6'-0", y=7'-9", y=9'-6" with LSCZ hangers

☐

Stringer bases secured to pavers with MSCZ anchors

☐

Both tread planks installed with pre-drilled Cortex face-screws; plugs flush

Section 8 — Far-edge step (v13/v14)

What you're doing

Installing the 11'-9" wide low step along the far (outer) edge of the deck, running from the right corner (x=23'-0") to the midpoint (x=11'-9"). A doubled 2×6 PT ribbon lag-bolts to the outer face of the far-side drop-board; five precast deck blocks on a gravel pad outside the patio footprint support the outboard edge of the ribbon.

Sequence

1. Prepare the gravel pad outside the patio: mark an 18"-wide strip from x=11'-9" to x=23'-0", running from the patio edge (y=16'-6") outward. Lay woven landscape fabric, then place and compact a 4" bed of #57 crushed stone. Level the pad across the full 11'-9" run.
2. Set five standard 12"×12"×8" notched precast deck blocks on the gravel pad, spaced approximately 28" o.c.: at x=23'-0", 20'-2", 17'-4", 14'-6", and 11'-9". Level each block; check level across all five with a 4-ft level or string line.
3. Cut the step ribbon inner ply (2×6 PT, 11'-9") from 12-ft stock. Seal the cut ends. Set it on the deck blocks temporarily to confirm fit; then mount it permanently to the outer face of the far-side drop-board/rim at the correct height so the step tread top lands at 6" AG. Lag-bolt with ½" × 3" HDG lags at ~12"

- o.c. in two rows (14–24 lags total depending on connection method — confirm with inspector before driving if through-bolting).
4. Install the outer ribbon ply (same 2×6 PT 11'-9"). Through-bolt or structural screw the two plies together at ~2 screws per lineal foot.
 5. Install the two MoistureShield Vision tread planks (5.4" × 11'-9", cut from 12-ft stock) on top of the doubled ribbon. Two planks side-by-side = 10.9" tread depth. Fasten with Cortex face-screws + plugs; pre-drill.
 6. Confirm the 6" AG step-top elevation with a level and tape.

Don't Forget — specific to this deck

- ■ **The five precast blocks are NOT frost-depth protected.** They will heave 1/4–1/2" each winter and resettle each spring. This is the accepted design — noted explicitly in the v15 BOM. Do not grout the blocks into the gravel pad; do not use hydraulic cement to anchor them. They are meant to be free to move. Check them for level each spring; tamp gravel and re-level if needed.
- ■ **The blocks are OUTSIDE the patio footprint (y > 16'-6").** They sit on native soil over gravel, not on concrete. Make sure your gravel pad starts at the patio edge (y=16'-6"), not inside it.
- ■ **The 3" exposed patio strip is visible and intentional.** The PF outer edge is at y=16'-3"; the patio edge is at y=16'-6". There is a 3" strip of bare concrete visible along the full outer edge. The right half (x=11'-9" to x=23'-0") is partially screened by the step assembly; the left half (x=0' to x=11'-9") is open and visible. This is not a framing error. The concrete patio drains away from the house; rain off the PF edge hits this strip and flows outward. No structural action needed.
- ■ **Confirm the ribbon-to-rim connection method with the inspector.** IRC interpretation for drop-mounted ribbons varies. Some inspectors want through-bolts rather than lags. Flag this at the pre-construction meeting so you're not stuck mid-installation.

Inspection / quality check before moving on

☐

Gravel pad level and compacted; landscape fabric in place

☐

All five precast blocks set level; checked with string line

☐

Ribbon fastened to far-side rim at correct height; step tread top at 6" AG

☐

Both tread planks installed with pre-drilled Cortex face-screws; plugs flush

☐

Ribbon connection method confirmed with inspector (lags vs. through-bolts)

Section 9 — Final inspection and closeout

What you're doing

Getting the final inspection signed off, cleaning the deck surface, and documenting the build for the homeowner's records.

Sequence

1. Do a full walk-through before calling for final inspection:
 - Confirm every Cortex plug is seated flush; replace any that are proud or loose.
 - Check PF miter corners for any gap; if there's movement from seasonal MC, address now.
 - Confirm all fascia trim screws are driven; no loose fascia panels.
 - Check all stair connections (both stairs) for racking or movement.
 - Check the far-edge step blocks for level one more time.
2. Call for final inspection. Bring the v15 BOM and permit documents.
3. After inspection sign-off, apply MoistureShield's recommended composite deck cleaner (Defy Composite Deck Cleaner or equivalent) per the product label. This removes construction residue and prepares the surface.
4. Document fastener torque for the through-bolt clusters at the three ledger connection points and the pier saddle bolts. Write these values in the homeowner's maintenance file (or in the Notes section at the back of this document).
5. Write down the MoistureShield Vision colorway name and the Cortex plug part number so any future replacement materials can be ordered to match.
6. Review the seasonal maintenance items with the homeowner:
 - Check far-edge step block level each spring; tamp gravel and re-level as needed.
 - Inspect ledger flashing and Z-flashing annually; no caulk to re-apply at top leg.
 - Inspect joist tape for any lift at edge seams; re-adhere with additional tape if lifting is found.

Don't Forget — specific to this deck

-  **The far-edge step's seasonal heave is documented, not a defect.** If the step rides ¼" high in March and settles back in May, that's normal. Do not fill, grout, or anchor it differently in response.
-  **No guardrail is required by IRC R312 because this deck is ≤ 12" AG throughout.** The West Hartford inspector should confirm this interpretation; don't let a well-meaning neighbor or subcontractor add a rail that isn't required unless the homeowner wants one aesthetically.

Inspection / quality check before moving on

☐

Final inspection passed

☐

- ☐ Deck surface cleaned with composite cleaner
- ☐ Through-bolt torque values documented
- ☐ Colorway + Cortex part numbers recorded
- ☐ Seasonal maintenance items reviewed with homeowner

Appendix A — Critical dimension lookup

All y-coordinates measured from the house wall face. All x-coordinates measured from the left end of the deck (x=0' at left girder / left corner).

Element	Dimension
Ledger top elevation	1.75" AG (= 12.0" – 1.0" decking – 9.25" joist)
Top of all framing members	11.0" AG
Top of MoistureShield Vision decking	12.0" AG (hard ceiling, non-negotiable)
Pier middle row (y)	8'-1.5" from house
Pier outer row (y)	16'-0" from house
Pier x-positions	x = 0', 11'-6", 23'-0"
Far-side joist rim (y)	16'-0" (framing outer edge)
Far-side drop-board + fascia outer face	~16'-0.67" (16'-0" rim + 0.67" fascia)
PF far-side outer edge	~16'-3" (rim outer face + 3" cantilever)
Patio outer edge	16'-6" from house
Exposed concrete strip (far edge)	3" (from PF outer face to patio edge)
Left girder (x)	x = 0'
Mid girder (x)	x = 11'-6"
Right girder (x)	x = 23'-0"
Left-edge stair opening (y)	y = 6'-0" to y = 9'-6" from house
Far-edge step run (x)	x = 11'-9" to x = 23'-0" (right half of deck)
Far-edge step ribbon y-position	16'-0" (same as rim outer face)
Step tread top elevation (both stairs)	6.0" AG
Step tread depth (both stairs)	10.8–10.9" (2 × 5.4" MoistureShield Vision)
Joist spacing	12" o.c.
Field blocking rows (x)	x = 5'-9" and x = 17'-3"
Field plank cut length	15'-7.2" (187.2") from 16-ft stock
Field plank count	49

Appendix B — “This deck specifically” cheat sheet

Print this page. Read it on Day 1 before picking up a tool. This is the list of everything about this deck that differs from what an experienced framer would default to.

1. Joists run PARALLEL to the house, not perpendicular. The three girders run from the house into the yard (perpendicular). The joists span across the yard parallel to the house. The decking runs perpendicular to the house (into the yard). If in doubt: girders go deep, joists go wide.

2. Only 6 piers in two rows — no house-side row. Middle row at $y=8'-1.5''$. Outer row at $y=16'-0''$. No piers within 12" of the house. The ledger carries the girder house-end loads at the three HU210-3 hanger points. Do not install any additional piers.

3. Outer pier row is at $y=16'-0''$, not $y=16'-3''$. If any previous mark on the patio says $y=16'-3''$ (a v13 mark), erase it and re-mark at $y=16'-0''$.

4. Ledger fastener density is non-uniform by design. Three through-bolt clusters (6 bolts each) at $x=0'$, $11'-6''$, $23'-0''$ — the girder-to-ledger connections. IRC lag pattern (18" o.c. staggered) between clusters. This is intentional load concentration at point-load zones.

5. Two different hanger types with two different nail/screw specs. - HU210-3 at ledger (3 locations, one per girder): read its spec sheet, use its specified fasteners. - **LUS210Z** in the field (75 total, joist-to-girder): read its spec sheet, use its specified fasteners. Do not swap. Do not substitute 16d commons for either.

6. Picture frame on three sides only, plus house-side flush. Far, left, and right: PF overhangs rim by 3" (or fascia by 2.33"). House side: PF butts the wall with $\frac{1}{4}''$ thermal gap, no fascia, no overhang. The PF cantilever is not adjustable — it is 3" by design.

7. Two separate, completely different stairs. Left-edge stair (v12): 42" wide, PT 2×10 cut stringers, three stringers, LSCZ hangers to left girder, precast pavers on gravel, $y=6'-0''$ to $9'-6''$. Far-edge step (v13/v14): 11'-9" long, doubled 2×6 PT ribbon, five precast deck blocks on gravel, attaches to far-side rim, runs from right corner to midpoint. These are independent assemblies. The left-side stair uses structural hangers; the far-edge step uses a ribbon. They have nothing in common except both land at 6" AG.

8. 3" exposed concrete strip at far edge — intentional, not a mistake. Deck PF outer edge is at $y=16'-3''$. Patio edge is at $y=16'-6''$. The 3" gap is bare concrete. Drains fine. Cosmetically acceptable as a narrow reveal. The homeowner is aware.

9. 12" o.c. joist spacing — not the typical 16". 49 field planks. ~70 face-mount hangers in the field (75 ordered with overage). This spec was chosen for stiffness under composite decking. Don't widen the spacing.

10. MoistureShield Vision fastener system is Tiger Claw TC-G + Cortex — not Trex Hideaway. The TC-G clip uses a #2 Robertson (square) bit. Buy extra bits. Pre-drill all face-screw locations in PF boards and tread planks. Cortex plug color must match the Vision colorway selected before any Cortex is ordered.

11. Composite decking spacing is not optional. MoistureShield Vision requires gapping per their installation guide (typically 1/8"). Butted planks will buckle in summer heat. Use spacer gauges throughout.

12. Joist tape goes on ALL framing tops — joists, girders, AND ledger. Not just the joists. Apply after framing inspection passes and before the first plank is set.

Appendix C — Inspection points

Stage	When to call	What inspector typically checks
Pier/ footing	After all 6 Pylex piers are driven and grout collars are placed; before any girder bears on them	Pier tip depth past 42" frost line; pier location within patio footprint; saddle elevation
Framing pre- decking	After all framing is complete (ledger, girders, joists, blocking, drop-boards, lateral ties); after LSCZ and HU210-3 hangers are nailed; BEFORE joist tape	Ledger connection (hanger + bolt pattern); through-bolt back-blocking; hanger nail spec compliance; joist spacing; DTT2Z placement; stair stringer hangers (LSCZ)
Post- decking	After decking (field planks + PF), fascia, and both stair assemblies are complete	Decking gap consistency; PF overhang; fascia fastening; stair tread height and depth; landing condition
Final	Project complete, cleaned	Overall condition; any remaining punch-list items; no guardrail required at $\leq 12"$ AG (confirm with inspector)

A note on timing: In West Hartford, schedule inspections 2–3 business days in advance. The framing inspection is the most critical for sequencing — don't apply joist tape, don't order your decking delivery, until this inspection is scheduled. Getting surprised by a 5-day inspector backlog with bare framing sitting in the rain is avoidable.

End of v15 Build Instructions. Prepared by Hollis — Home-Improvement Concierge / Owner's Advocate. Cross-reference with v15_bill_of_materials.md for exact quantities and ordering notes. If anything in these instructions conflicts with the West Hartford inspector's interpretation of the IRC, defer to the inspector.